UNIVERSIDAD DEL BÍO-BÍO

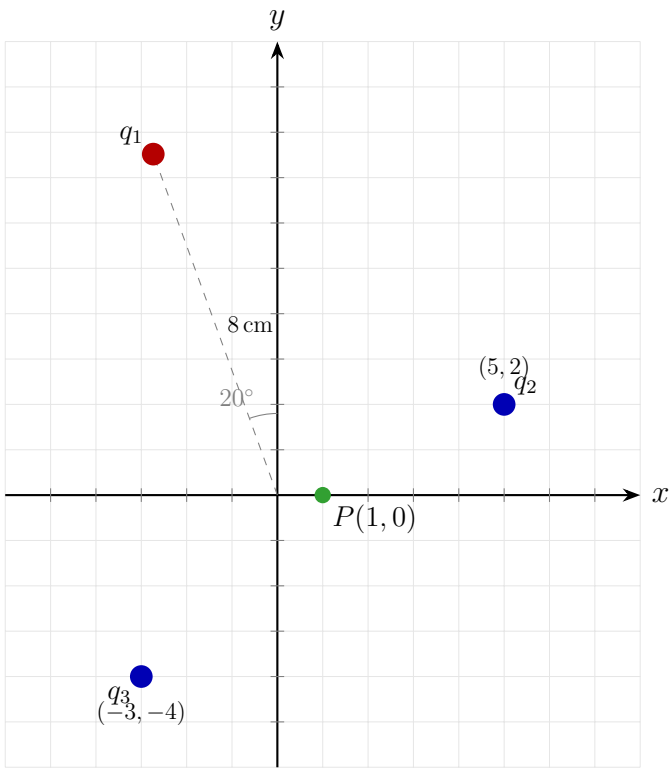
Facultad de Ciencias – Departamento de Física

Solución — Certamen 1 de Electromagnetismo

18 de octubre de 2023

Problema 1: Tres Cargas Puntuales (29 puntos)

Tres cargas puntuales: $q_1 = 11{,}8$ nC, $q_2 = -31{,}5$ nC, $q_3 = -10{,}3$ nC. La carga q_1 se encuentra a 8,0 cm del origen, 20° del eje y; q_2 en $(5, 2)$ cm; q_3 en $(-3, -4)$ cm.



Datos Numéricos

Posición de q_1 (a 8,0 cm del origen, 20° del eje y hacia el 2do cuadrante):

$$x_1 = -8{,}0\sin(20°) = -2{,}74 \text{ cm} = -0{,}0274 \text{ m}$$
$$y_1 = +8{,}0\cos(20°) = +7{,}52 \text{ cm} \approx 0{,}075 \text{ m}$$

$$\vec{r}_1 = (-0{,}0274\,\hat{\imath} + 0{,}075\,\hat{\jmath}) \text{ m} \qquad q_1 = +11{,}8 \times 10^{-9} \text{ C}$$
$$\vec{r}_2 = (+0{,}05\,\hat{\imath} + 0{,}02\,\hat{\jmath}) \text{ m} \qquad q_2 = -31{,}5 \times 10^{-9} \text{ C}$$
$$\vec{r}_3 = (-0{,}03\,\hat{\imath} - 0{,}04\,\hat{\jmath}) \text{ m} \qquad q_3 = -10{,}3 \times 10^{-9} \text{ C}$$
$$K = 9 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$$

Parte (a): Fuerza eléctrica neta sobre q_2 — 16 puntos

$$\vec{F}_2 = \vec{F}_{21} + \vec{F}_{23} \qquad \text{donde} \qquad \vec{F}_{ij} = K\frac{q_i\,q_j}{\|\vec{r}_{ij}\|^3}\,\vec{r}_{ij}, \quad \vec{r}_{ij} = \vec{r}_j - \vec{r}_i$$

Análisis Cualitativo

- $q_1 > 0$, $q_2 < 0 \Rightarrow$ **atracción**: $\vec{F}_{21}$ apunta de q_2 hacia q_1.
- $q_3 < 0$, $q_2 < 0 \Rightarrow$ **repulsión**: $\vec{F}_{23}$ apunta de q_3 alejándose (dirección $q_3 \to q_2$).

Fuerza $\vec{F}_{21}$ (de q_1 sobre q_2)

$$\vec{r}_{21} = \vec{r}_2 - \vec{r}_1 = \big(0{,}05 - (-0{,}0274)\big)\,\hat{\imath} + \big(0{,}02 - 0{,}075\big)\,\hat{\jmath} = 0{,}0774\,\hat{\imath} - 0{,}055\,\hat{\jmath} \text{ m}$$
$$\|\vec{r}_{21}\| = \sqrt{(0{,}0774)^2 + (0{,}055)^2} \approx 0{,}095 \text{ m}$$

$$\vec{F}_{21} = K\frac{q_1\,q_2}{\|\vec{r}_{21}\|^3}\,\vec{r}_{21} = \frac{9 \times 10^9 \cdot (11{,}8 \times 10^{-9}) \cdot (-31{,}5 \times 10^{-9})}{(0{,}095)^3}\,(0{,}0774\,\hat{\imath} - 0{,}055\,\hat{\jmath})$$
$$= \frac{-3{,}345 \times 10^{-6}}{8{,}574 \times 10^{-4}}\,(0{,}0774\,\hat{\imath} - 0{,}055\,\hat{\jmath}) = -3{,}902 \times 10^{-3}\,(0{,}0774\,\hat{\imath} - 0{,}055\,\hat{\jmath})$$
$$\vec{F}_{21} = \big(-3{,}02\,\hat{\imath} + 2{,}15\,\hat{\jmath}\big) \times 10^{-4} \text{ N}$$

Fuerza $\vec{F}_{23}$ (de q_3 sobre q_2)

$$\vec{r}_{23} = \vec{r}_2 - \vec{r}_3 = \big(0{,}05 - (-0{,}03)\big)\,\hat{\imath} + \big(0{,}02 - (-0{,}04)\big)\,\hat{\jmath} = 0{,}08\,\hat{\imath} + 0{,}06\,\hat{\jmath}\ \text{m}$$
$$\|\vec{r}_{23}\| = \sqrt{(0{,}08)^2 + (0{,}06)^2} = 0{,}10\ \text{m}$$

$$\vec{F}_{23} = K\frac{q_3\,q_2}{\|\vec{r}_{23}\|^3}\,\vec{r}_{23} = \frac{9\times10^9\cdot(-10{,}3\times10^{-9})\cdot(-31{,}5\times10^{-9})}{(0{,}10)^3}\,(0{,}08\,\hat{\imath} + 0{,}06\,\hat{\jmath})$$
$$= \frac{+2{,}920\times10^{-6}}{1{,}0\times10^{-3}}\,(0{,}08\,\hat{\imath} + 0{,}06\,\hat{\jmath}) = 2{,}920\times10^{-3}\,(0{,}08\,\hat{\imath} + 0{,}06\,\hat{\jmath})$$
$$\vec{F}_{23} = \big(+2{,}34\,\hat{\imath} + 1{,}75\,\hat{\jmath}\big)\times10^{-4}\ \text{N}$$

Superposición

$$F_{2x} = (-3{,}02 + 2{,}34)\times10^{-4} = -0{,}68\times10^{-4}\ \text{N}$$
$$F_{2y} = (+2{,}15 + 1{,}75)\times10^{-4} = +3{,}90\times10^{-4}\ \text{N}$$

$$\boxed{\vec{F}_2 = \big(-6{,}68\times10^{-5}\,\hat{\imath} + 3{,}90\times10^{-4}\,\hat{\jmath}\big)\ \text{N} \qquad \|\vec{F}_2\| \approx 3{,}96\times10^{-4}\ \text{N}}$$

Parte (b): Campo eléctrico neto en $P(1,0)$ cm — 13 puntos

El punto campo es $\vec{r}_P = 0{,}01\,\hat{\imath}$ m. Se aplica superposición:

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3, \qquad \vec{E}_i = K\frac{q_i}{\|\vec{r}_P - \vec{r}_i\|^3}\,(\vec{r}_P - \vec{r}_i)$$

Campo $\vec{E}_1$ (de $q_1 = +11{,}8$ nC)

$$\vec{r}_P - \vec{r}_1 = \big(0{,}01 - (-0{,}0274)\big)\,\hat{\imath} + (0 - 0{,}075)\,\hat{\jmath} = 0{,}037\,\hat{\imath} - 0{,}075\,\hat{\jmath}\ \text{m} \qquad \|\cdot\| \approx 0{,}084\ \text{m}$$

$$\begin{aligned}\vec{E}_1 &= \frac{9\times10^9\cdot 11{,}8\times10^{-9}}{(0{,}084)^3}\,(0{,}037\,\hat{\imath} - 0{,}075\,\hat{\jmath})\\ &= 1{,}790\times10^5\,(0{,}037\,\hat{\imath} - 0{,}075\,\hat{\jmath})\\ &= \big(6{,}62\times10^3\,\hat{\imath} - 1{,}34\times10^4\,\hat{\jmath}\big)\ \text{N/C}\end{aligned}$$

Campo $\vec{E}_2$ (de $q_2 = -31{,}5$ nC)

$$\vec{r}_P - \vec{r}_2 = (0{,}01 - 0{,}05)\,\hat{\imath} + (0 - 0{,}02)\,\hat{\jmath} = -0{,}04\,\hat{\imath} - 0{,}02\,\hat{\jmath}\ \text{m} \qquad \|\cdot\| \approx 0{,}045\ \text{m}$$

$$\begin{aligned}\vec{E}_2 &= \frac{9\times10^9\cdot(-31{,}5\times10^{-9})}{(0{,}045)^3}\,(-0{,}04\,\hat{\imath} - 0{,}02\,\hat{\jmath})\\ &= (-3{,}111\times10^6)\,(-0{,}04\,\hat{\imath} - 0{,}02\,\hat{\jmath})\\ &= \big(1{,}244\times10^5\,\hat{\imath} + 6{,}222\times10^4\,\hat{\jmath}\big)\ \text{N/C}\end{aligned}$$

Campo $\vec{E}_3$ (de $q_3 = -10{,}3$ nC)

$$\vec{r}_P - \vec{r}_3 = \big(0{,}01 - (-0{,}03)\big)\,\hat{\imath} + \big(0 - (-0{,}04)\big)\,\hat{\jmath} = 0{,}04\,\hat{\imath} + 0{,}04\,\hat{\jmath}\ \text{m} \qquad \|\cdot\| \approx 0{,}057\ \text{m}$$

$$\begin{aligned}\vec{E}_3 &= \frac{9\times10^9\cdot(-10{,}3\times10^{-9})}{(0{,}057)^3}\,(0{,}04\,\hat{\imath} + 0{,}04\,\hat{\jmath})\\ &= (-5{,}008\times10^5)\,(0{,}04\,\hat{\imath} + 0{,}04\,\hat{\jmath})\\ &= \big(-2{,}00\times10^4\,\hat{\imath} - 2{,}00\times10^4\,\hat{\jmath}\big)\ \text{N/C}\end{aligned}$$

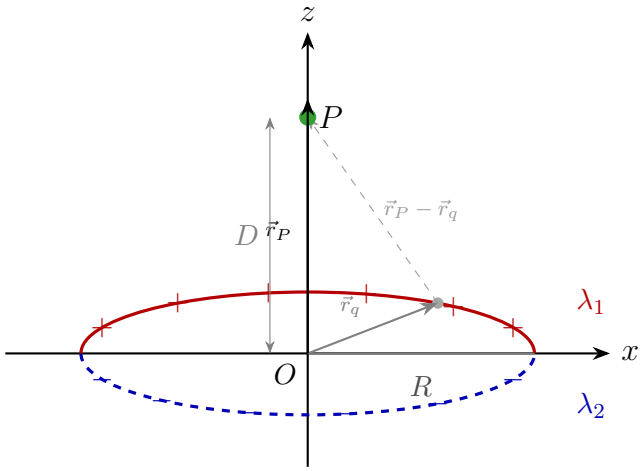
Superposición

$$E_x = 6{,}62 \times 10^3 + 1{,}244 \times 10^5 - 2{,}00 \times 10^4 = 1{,}11 \times 10^5 \text{ N/C}$$
$$E_y = -1{,}34 \times 10^4 + 6{,}222 \times 10^4 - 2{,}00 \times 10^4 = 2{,}88 \times 10^4 \text{ N/C}$$

$$\boxed{\vec{E} = \left(1{,}11 \times 10^5\,\hat{\imath} + 2{,}88 \times 10^4\,\hat{\jmath}\right) \text{ N/C} \qquad \|\vec{E}\| \approx 1{,}15 \times 10^5 \text{ N/C}}$$

Problema 2: Anillo con Dos Densidades Lineales (30 puntos)

Anillo de radio $R = 14{,}0$ cm en el plano xy: semianillo superior ($0 \leq \theta \leq \pi$) con $\lambda_1 = +3{,}6$ nC/m; semianillo inferior ($\pi \leq \theta \leq 2\pi$) con $\lambda_2 = -5{,}8$ nC/m. El punto P se ubica en el eje del anillo ($\hat{k}$) a distancia $D = 8{,}0$ cm de su centro.



Datos Numéricos

$$R = 0{,}14 \text{ m} \qquad D = 0{,}08 \text{ m} \qquad K = 9 \times 10^9 \text{ N·m}^2/\text{C}^2$$
$$\lambda_1 = +3{,}6 \times 10^{-9} \text{ C/m} \qquad \lambda_2 = -5{,}8 \times 10^{-9} \text{ C/m}$$

Parte (a): Campo eléctrico en P — 22 puntos

Configuración del sistema

$$\vec{r}_P = D\,\hat{k} \quad \text{(vector al punto campo)}$$
$$\vec{r}_q = R\cos\theta\,\hat{\imath} + R\sin\theta\,\hat{\jmath} \quad \text{(vector al elemento fuente)}$$
$$dq = \lambda\,dl = \lambda R\,d\theta$$

Vector de separación del elemento fuente al punto campo:

$$\vec{r}_P - \vec{r}_q = -R\cos\theta\,\hat{\imath} - R\sin\theta\,\hat{\jmath} + D\,\hat{k} \qquad \|\vec{r}_P - \vec{r}_q\| = \sqrt{R^2 + D^2} = \text{cte.}$$

Campo diferencial:

$$d\vec{E} = K\frac{\lambda R\,d\theta}{(R^2 + D^2)^{3/2}}\left(-R\cos\theta\,\hat{\imath} - R\sin\theta\,\hat{\jmath} + D\,\hat{k}\right)$$

Contribución del semianillo 1 ($0 \leq \theta \leq \pi$, $\lambda = \lambda_1$)

$$\vec{E}_1 = \frac{K\lambda_1 R}{(R^2+D^2)^{3/2}}\left(-R\int_0^{\pi}\cos\theta\, d\theta\ \hat{\imath} - R\int_0^{\pi}\sin\theta\, d\theta\ \hat{\jmath} + D\int_0^{\pi} d\theta\ \hat{k}\right)$$

Evaluando las integrales:

$$\int_0^{\pi}\cos\theta\, d\theta = [\sin\theta]_0^{\pi} = 0$$
$$\int_0^{\pi}\sin\theta\, d\theta = [-\cos\theta]_0^{\pi} = -(-1)+(1) = 2$$
$$\int_0^{\pi} d\theta = \pi$$

$$\vec{E}_1 = \frac{K\lambda_1 R}{(R^2+D^2)^{3/2}}\left(-2R\,\hat{\jmath} + \pi D\,\hat{k}\right)$$

Factor numérico:

$$(R^2+D^2)^{3/2} = (0{,}14^2+0{,}08^2)^{3/2} = (0{,}026)^{3/2} = 0{,}026\times\sqrt{0{,}026} = 4{,}191\times10^{-3}\ \mathrm{m}^3$$

$$\frac{K\lambda_1 R}{(R^2+D^2)^{3/2}} = \frac{9\times10^9\times3{,}6\times10^{-9}\times0{,}14}{4{,}191\times10^{-3}} = \frac{4{,}536}{4{,}191\times10^{-3}} = 1082\ \frac{\mathrm{N}}{\mathrm{C{\cdot}m^2}}$$

$$\vec{E}_1 = 1082\times\left(-2\times0{,}14\,\hat{\jmath} + \pi\times0{,}08\,\hat{k}\right) = 1082\times\left(-0{,}28\,\hat{\jmath} + 0{,}2513\,\hat{k}\right)$$
$$\vec{E}_1 = -303{,}0\,\hat{\jmath} + 271{,}9\,\hat{k} \quad [\mathrm{N/C}]$$

Contribución del semianillo 2 ($\pi \leq \theta \leq 2\pi$, $\lambda = \lambda_2$)

$$\vec{E}_2 = \frac{K\lambda_2 R}{(R^2+D^2)^{3/2}}\left(-R\int_{\pi}^{2\pi}\cos\theta\, d\theta\ \hat{\imath} - R\int_{\pi}^{2\pi}\sin\theta\, d\theta\ \hat{\jmath} + D\int_{\pi}^{2\pi} d\theta\ \hat{k}\right)$$

Evaluando las integrales:

$$\int_{\pi}^{2\pi}\cos\theta\, d\theta = [\sin\theta]_{\pi}^{2\pi} = 0-0 = 0$$
$$\int_{\pi}^{2\pi}\sin\theta\, d\theta = [-\cos\theta]_{\pi}^{2\pi} = -\cos(2\pi)+\cos(\pi) = -1+(-1) = -2$$
$$\int_{\pi}^{2\pi} d\theta = \pi$$

$$\vec{E}_2 = \frac{K\lambda_2 R}{(R^2+D^2)^{3/2}}\left(+2R\,\hat{\jmath} + \pi D\,\hat{k}\right)$$

$$\frac{K\lambda_2 R}{(R^2+D^2)^{3/2}} = \frac{9\times10^9\times(-5{,}8\times10^{-9})\times0{,}14}{4{,}191\times10^{-3}} = \frac{-7{,}308}{4{,}191\times10^{-3}} = -1744\ \frac{\mathrm{N}}{\mathrm{C{\cdot}m^2}}$$

$$\vec{E}_2 = -1744\times\left(+2\times0{,}14\ \hat{\jmath} + \pi\times0{,}08\ \hat{k}\right) = -1744\times\left(0{,}28\ \hat{\jmath} + 0{,}2513\ \hat{k}\right)$$
$$\vec{E}_2 = -488{,}3\ \hat{\jmath} - 438{,}3\ \hat{k} \quad [\mathrm{N/C}]$$

Campo total en P

$$E_j = -303{,}0 + (-488{,}3) = -791{,}3\ \mathrm{N/C}$$
$$E_k = +271{,}9 + (-438{,}3) = -166{,}4\ \mathrm{N/C}$$

$$\boxed{\vec{E} = \left(-791\ \hat{\jmath} - 166\ \hat{k}\right)\ \mathrm{N/C} \qquad \|\vec{E}\| = \sqrt{791^2+166^2} \approx 810\ \mathrm{N/C}}$$

Nota: la componente $\hat{\imath}$ es nula porque las contribuciones de $\cos\theta$ se cancelan al integrar sobre cada semianillo completo.

Parte (b): Fuerza sobre $q = -88{,}0$ nC en P — 8 puntos

$$\vec{F} = q\,\vec{E} = (-88{,}0\times10^{-9})\left(-791\ \hat{\jmath} - 166\ \hat{k}\right)$$
$$= 69{,}6\times10^{-6}\ \hat{\jmath} + 14{,}6\times10^{-6}\ \hat{k}$$

$$\boxed{\vec{F} = \left(6{,}96\times10^{-5}\ \hat{\jmath} + 1{,}46\times10^{-5}\ \hat{k}\right)\ \mathrm{N}}$$

La carga negativa experimenta una fuerza opuesta al campo: hacia $+\hat{\jmath}$ y $+\hat{k}$.